Controller Microarchitecture

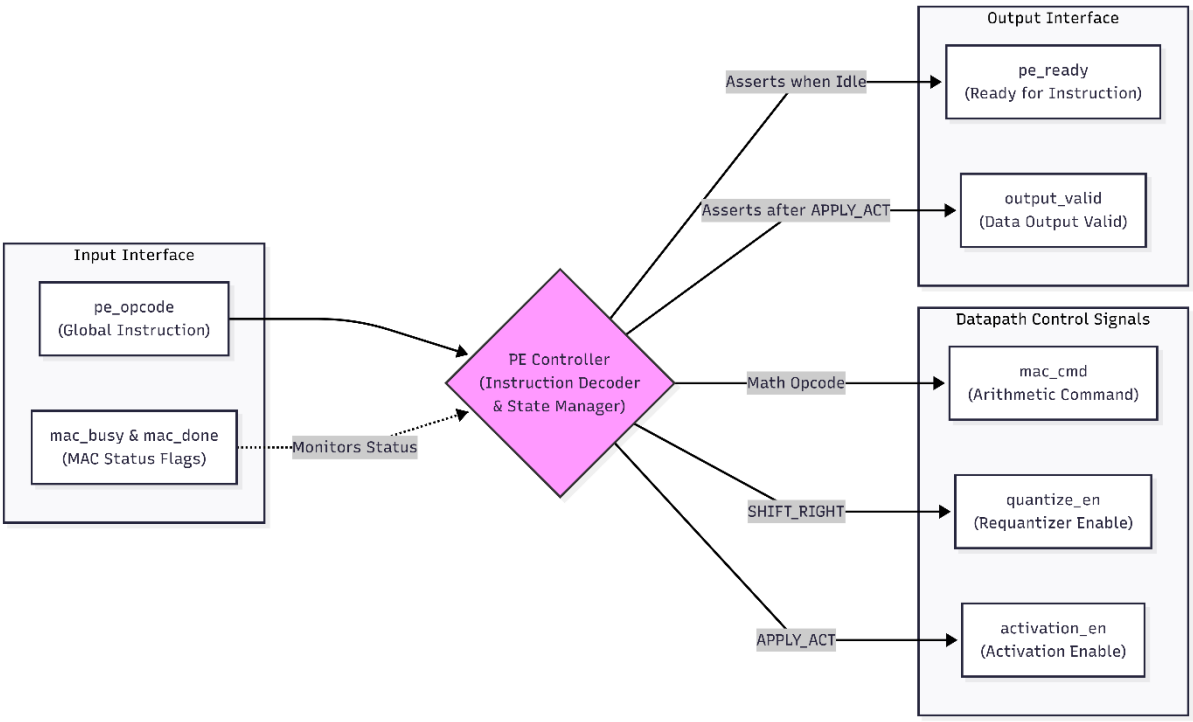

Figure 1. Block-level microarchitecture.

High-Level Overview

The PE_Controller serves as the centralized instruction decoder and state manager for the PE. It receives a 5-bit instruction (pe_opcode) from the top-level module, decodes the requested operation, and asserts the control and enable signals to the underlying datapath modules (MAC, Requantizer, and Activation). By isolating the control and sequencing logic within this dedicated module, the top-level PE architecture remains strictly structural, acting solely as a routing fabric to interconnect the instantiated sub-modules.

Interface

Port Name	Direction	Width	Description
clk	Input	1 bit	Main system clock.
rst	Input	1 bit	Active-low asynchronous reset.
pe_opcode	Input	5-bit bus	The main instruction fed into the PE from the top-level inputs. Supports up to 32 distinct operations.
mac_busy	Input	1 bit	Status flag from the MAC unit indicating an ongoing multi-cycle operation.
mac_done	Input	1 bit	Status flag from the MAC unit indicating a multi-cycle operation has just completed.
mac_cmd	Output	5-bit bus	The decoded arithmetic operational code routed directly to the MAC unit.
quantize_en	Output	1 bit	Synchronous enable trigger routed to the Requantizer.
activation_en	Output	1 bit	Global enable trigger routed to the Activation block.
pe_ready	Output	1 bit	Flow-control flag routed to the top-level output. It indicates to the instruction source that the PE has finished a multi-cycle operation and is idle, ready to accept a new instruction.
output_valid	Output	1 bit	Asserted high for one clock cycle exactly two clock cycles after the APPLY_ACT operation is issued. Indicates to the external system that the final 8-bit activation result is stable and valid on the PE's output bus.

Opcodes

Opcode	Mnemonic	Full Name	Description
00000	NOP	No Operation	The controller takes no action. All datapath modules hold their current state and wait.
00001	RST_ACC	Reset Accumulator	Routed directly to the MAC unit. Clears the internal 64-bit signed accumulator to zero in preparation for a new calculation.
00010	MAC4	Multiply and Accumulate INT4	Routed directly to the MAC unit. The datapath treats the 32-bit inputs as eight parallel 4-bit signed integers and accumulates the result.
00011	MAC8	Multiply and Accumulate INT8	Routed directly to the MAC unit. The datapath treats the 32-bit inputs as four parallel 8-bit signed integers and accumulates the result.
00100	ADD_BIAS	Add Bias	Routed directly to the MAC unit. Adds the 32-bit bias value from the operand bus directly into the active accumulator.
00101	MULT32	32x32 Multiplication	Routed directly to the MAC unit to trigger a multi-cycle multiplication. The controller pulls pe_ready low to stall the system until the MAC finishes.
00110	SHIFT_RIGHT	Shift Right (Requantize)	The controller intercepts this command. Once the MAC is idle, the controller pulses quantize_en. The Requantizer scales the 64-bit accumulator down to an 8-bit value using the static shift amount register.
00111	APPLY_ACT	Apply Activation	The controller intercepts this command and pulses activation_en. The Activation block evaluates the scaled 8-bit value using the static activation selection register.
01000 to 11111	RSV	Reserved	Unused binary encodings left open for future feature expansion.

Instruction Decode and Routing

The controller sequentially decodes the incoming 5-bit pe_opcode to control datapath execution. All output control signals are registered at the clock edge, introducing a strict one-cycle latency between receiving an instruction and firing the datapath signals:

- **Arithmetic Operations (MAC):** For operations requiring the arithmetic core (MAC4, MAC8, MULT32), the controller forwards the specific 5-bit command to the MAC unit via the mac_cmd port. During these active math cycles, the quantize_en and activation_en signals are driven low to disable downstream logic and prevent unintended combinational switching.
- **Scaling and Activation:** The Requantizer and Activation modules operate as conditional execution blocks. When a scaling (SHIFT_RIGHT) or non-linear mapping (APPLY_ACT) instruction is decoded, the controller does not transmit a multi-bit operational code. Instead, it asserts a localized 1-bit enable signal (quantize_en or activation_en) at the correct clock cycle to trigger the respective operation, relying on the pre-configured static inputs to define the specific function behavior.

Synchronization and State Management

The PE_Controller relies on the status flags generated by the MAC unit (mac_busy and mac_done) to track the execution state of a multi-cycle operation and ensure synchronous data flow across the PE interface.

When the decoded pe_opcode initiates a multi-cycle operation, the controller immediately deasserts its pe_ready output. This drives a "stall" signal to the external interface, indicating that the PE cannot accept new instructions. The controller then actively monitors the mac_done input port. Upon detecting a high pulse on mac_done, the controller registers the completion of the arithmetic operation, and immediately reasserts the pe_ready flag high.

Once pe_ready is high, the controller takes no autonomous action and waits for the external system to drive the next intended instruction onto the pe_opcode pins.